The Blind Spot Paradox

When a self-repairing model destroys the evidence its own monitor needs

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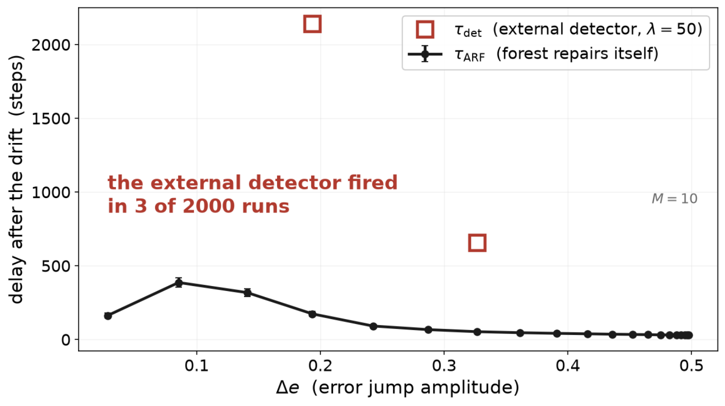
Market finance: who tells you to retrain?

- A classifier predicts the sign of tomorrow's CAC 40 return.
- A macro shock reverses momentum: $P(y \mid X)$ changes while X still looks the same.
- Nobody retrains on intuition. A drift detector watches the error stream and triggers the retraining, the audit, the pause.

So the detector is the part that must not fail.

The classifier erases its own alarm

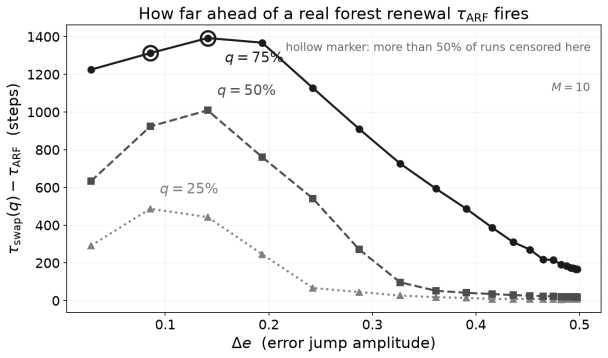


Authors' experiment R2, re-run locally. $\lambda = 50$, 2 000 runs, 20 amplitudes $\times$ 100 seeds.

Two literatures, one missing clock

- **CUSUM** (Page, 1954): the alarm delay grows as the threshold divided by the size of the jump, $\tau^*_{\mathrm{det}} = \lambda/(\Delta e - \delta_P)$.
- **Adaptive Random Forest** (Gomes et al., 2017): evaluated on predictive accuracy alone.
- **Gama et al. 2014, Lu et al. 2018**, the two reference surveys: neither lists a measure of how long a model takes to recover.
- **Source paper**: on heteroscedastic streams, no admissible threshold exists. $\lambda \geq 15$ to avoid false alarms, $\lambda \leq 12.4$ to catch the drift.

We measured the missing clock



τ_{ARF} dates the first tree out of ten. A real forest renewal comes 6.0 to 14.3 times later.

Where we stand

- **Proved**: a deterministic certificate on the accumulated evidence, true run by run, with no distributional assumption.
- **Measured**: a violent shock leaves no more evidence than a slow one.
- **Next, question D**: instrument the forest and the detector on one common time axis, and turn this into a decision rule.

Thank you.